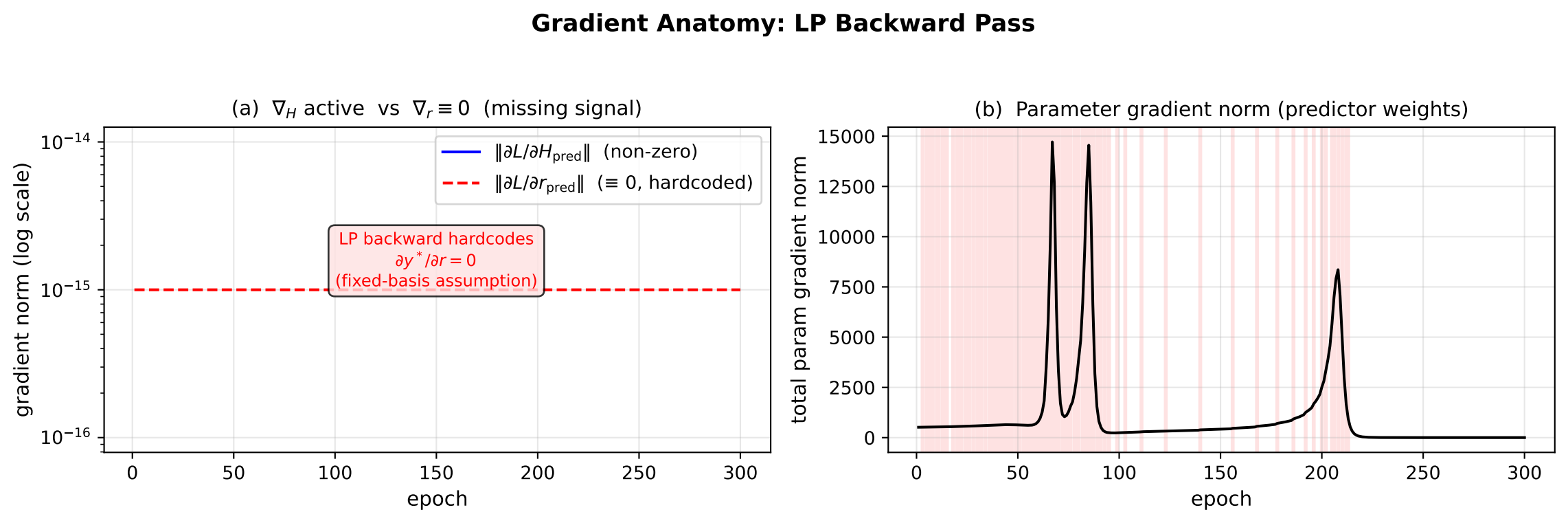
Gradient Anatomy: LP Backward Pass
(a) ∇_H active vs $\nabla_r \equiv 0$ (missing signal)
gradient norm (log scale)
10^{-14}
10^{-15}
10^{-16}
$\|\partial L/\partial H_{\text{pred}}\|$ (non-zero)
$\|\partial L/\partial r_{\text{pred}}\|$ (≡ 0, hardcoded)
LP backward hardcodes
$\partial y^*/\partial r = 0$
(fixed-basis assumption)
0
50
100
150
200
250
300
epoch
(b) Parameter gradient norm (predictor weights)
total param gradient norm
15000
12500
10000
7500
5000
2500
0
0
50
100
150
200
250
300
epoch